

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 14

What a Model Is and Why It Works

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. A model is described as “wrong but useful” (Box). Restate this using the language of structure-preserving maps. What exactly is “wrong”? What exactly is “useful”?

2. You fit a model and it performs well on training data but poorly on test data. Diagnose this in information-theoretic terms (from Chapter 12) and in structural terms (from this chapter). Are the diagnoses related?

3. A physicist and a biologist build models of the same phenomenon (e.g., fluid flow in blood vessels). The physicist’s model is a PDE. The biologist’s model is a statistical correlation. Both “work.” Are they capturing the same structure? If not, what is each one capturing?

4. Someone says: “With enough parameters, any model can fit any data.” Is this true? If so, why is it a *problem* rather than a *solution*?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
